

M15 Independent Practice

Fresh work from blank state

OPTION B | 16 LESSON ROUTES

Protected Review stays closed until attempt is saved.

DE15-01 - Cloud fundamentals for data engineers

INDEPENDENT | Teaching Book p6 | route page 2

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Create the same map for a clinic ingestion case without copying the guided layout.

Validation

Produce architecture map; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-01/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-01/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-02 - Identity, IAM and least privilege

INDEPENDENT | Teaching Book p9 | route page 3

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Build a fresh matrix for a contractor who can monitor runs but must not edit production data.

Validation

Produce least-privilege permission matrix; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-02/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-02/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-03 - Azure Storage concepts and OneLake

INDEPENDENT | Teaching Book p12 | route page 4

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Evaluate a fresh source where the producer controls lifecycle and you need near-current reads.

Validation

Produce copy vs shortcut decision; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-03/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-03/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-04 - Fabric workspace and item model

INDEPENDENT | Teaching Book p15 | route page 5

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Create a fresh inventory for a support-analytics workload.

Validation

Produce workspace/item inventory; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-04/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-04/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-05 - OneLake paths, files, tables and shortcuts

INDEPENDENT | Teaching Book p18 | route page 6

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Design a changed path map for a vendor-owned external data source.

Validation

Produce OneLake path and shortcut map; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE `04\_PRACTICE\_FILES/lesson\_workspaces/DE15-05/attempt.md` and evidence under `04\_PRACTICE\_FILES/lesson\_workspaces/DE15-05/evidence/`. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-06 - Fabric Lakehouse and Delta tables

INDEPENDENT | Teaching Book p21 | route page 7

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Define the same contract for a fresh clinic-events table with explicit bad-row reasons.

Validation

Produce Lakehouse/Delta contract; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-06/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-06/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-07 - Fabric notebooks

INDEPENDENT | Teaching Book p24 | route page 8

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Design a fresh notebook flow for device telemetry without relying on hidden session state.

Validation

Produce rerunnable notebook flow; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-07/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-07/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-08 - Spark in Fabric

INDEPENDENT | Teaching Book p27 | route page 9

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Create a fresh checklist for moving a local Spark/Delta job to Fabric without assuming preview features.

Validation

Produce Fabric runtime compatibility checklist; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-08/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-08/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-09 - Fabric Data Pipelines

INDEPENDENT | Teaching Book p30 | route page 10

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Create a fresh graph where validation fails and prove publish is blocked.

Validation

Produce pipeline dependency and failure graph; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-09/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-09/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-10 - SQL analytics endpoint and warehouse choice

INDEPENDENT | Teaching Book p33 | route page 11

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Choose for a fresh finance-serving case and defend the alternative you rejected.

Validation

Produce SQL endpoint vs Warehouse choice; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE `04\_PRACTICE\_FILES/lesson\_workspaces/DE15-10/attempt.md` and evidence under `04\_PRACTICE\_FILES/lesson\_workspaces/DE15-10/evidence/`. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-11 - Medallion architecture in Fabric

INDEPENDENT | Teaching Book p36 | route page 12

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Design a changed medallion flow for clinic appointments.

Validation

Produce medallion architecture; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-11/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-11/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-12 - Incremental loads and MERGE in cloud

INDEPENDENT | Teaching Book p39 | route page 13

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Design a changed incremental load where a failed publish must not advance its watermark.

Validation

Produce incremental MERGE state transition; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-12/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-12/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-13 - Permissions, connections and secrets

INDEPENDENT | Teaching Book p42 | route page 14

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Create a fresh cross-workspace connection plan without embedding credentials.

Validation

Produce permissions, connections and secrets; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-13/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-13/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-14 - Monitoring, reliability and cost awareness

INDEPENDENT | Teaching Book p45 | route page 15

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Create a fresh runbook for a slow pipeline whose output totals still reconcile.

Validation

Produce monitoring/reliability/cost runbook; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-14/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-14/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-15 - Git, deployment and environment separation

INDEPENDENT | Teaching Book p48 | route page 16

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Introduce a changed test environment with different IDs and revise promotion.

Validation

Produce Git/deployment/environment separation; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-15/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-15/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.

DE15-16 - End-to-end Fabric data engineering case

INDEPENDENT | Teaching Book p51 | route page 17

Close the guided answer

Start from a blank page/workspace. Do not copy the teacher wording.

Your independent task

Produce the complete clinic changed-case architecture/evidence plan from a blank template.

Validation

Produce end-to-end Fabric evidence index; independently check the important state/count/permission/recovery claim.

Evidence to CREATE

CREATE ``04_PRACTICE_FILES/lesson_workspaces/DE15-16/attempt.md`` and evidence under ``04_PRACTICE_FILES/lesson_workspaces/DE15-16/evidence/``. Preserve errors and uncertainty.

Move-on rule

Independent result is reproducible, explainable without the guided page, correctly labeled LIVE/OFFLINE/SIMULATOR, and has no unresolved critical failure.